Tiling and Coloring

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Problem 1

Is it possible to cover a 10×10 grid using $T - tetrominoes$?

Problem 2

Is it possible to cover a 10×10 grid using 1×4 and 4×1 tetrominoes?

Problem 3

A grid (not necessarily rectangular) is covered using A of 2×2 tetrominoes and B of 1×4 tetrominoes. Is it possible to cover the same grid using $A - 1$ of 2×2 tetrominoes and $B + 1$ of 1×4 tetrominoes?

Problem 4

A corner cell of a $(2n - 1) \times (2n - 1)$ board is deleted. For which n is it possible to cover the rest of the grid using the same number of 2×1 and 1×2 dominoes?

Problem 5

There are 81 Bihan-pukas standing in each cell of a 9×9 grid. During the great Bihan movement, all the Bihan-pukas simultaneously move to a horizantally adjacent square. At least how many cells will be empty after the great Bihan movement?

Problem 6

All the cells of a 5×5 grid is colored white. G8 changes the color of one cell into black. Now, NNT can choose a $a \times a$ square in the grid ($a > 1$) and swap the color of all the cells in the square. NNT wins if he can turn the whole board white (He lowk racist). What initial choice by G8 ensures that NNT will lose?

Problem 7

Can you make a subset of the set of positive odd integers, say S such that $|S| > 1$ and for odd x, y

- If $x, y \in S$ then $xy \in S$
- If $x \in S$ and $y \notin S$ then $xy \notin S$
- If $x, y \notin S$ then $xy \in S$

Problem 8

All 5 vertices of a pentagon are lattice points and the side lengths are integers. Prove that the perimeter of the pentagon is even.

Problem 9

Consider three vertices $A = (0, 0), B = (0, 1)$ and $C(1, 0)$ in a set S.
Peak-Saad is allowed to choose points P, Q in S and get $R =$ the reflection of P over Q and then insert R in S.
Which lattice points can Peak-Saad reach using this?

Problem 10

The residue class modulo $2n$ is divided into n disjoint pairs $\{a_1, b_1\}, \{a_2, b_2\}, \{a_3, b_3\}...\{a_n, b_n\}$. Prove that there exists two pairs $\{a_i, b_i\}$ and $\{a_j, b_j\}$ such that $|a_i - b_i| \equiv |a_j - b_j| \pmod{2n}$ if $n \equiv 2, 3 \pmod 4$

Problem 11

15 rooks are placed on a 15×15 board such that they do not attack each other. At 11:56am of 15/3/26 all the rooks make a knight move. Prove that there are at least two rooks that attack each other after the movement.

Problem 12

NNT and G8 are playing a game on a 100×1000 board. There are 100 clack pawns in the 1000th row and 100 white pawns in the first row. They may not play optimally. What is the minimum number of pawns remaining after some finite amount of moves? (Pawns move and capture similar to regular chess.)

Note: Ami pss theke copy korinai but coincidentally almost shob prob okhane ase so sol okhanei pabe.